BCS-063	SYSTEM PROGRAMMING	L T P 3 0 0	3 Credits

Pre-requisites: None
Course Objectives:

CO1	Develop complex applications using asynchronous programming techniques.
CO2	Understand the basic operating system concepts.
CO3	Understand the basics of networking technologies and have good understanding of communication networking concepts and protocols.
CO4	Display proficiency in the use of programming languages like C.
CO5	Understand the fundamentals of concurrent programming.

Detailed Syllabus

MODULE-I

Evolution of Components Systems Programming, Assemblers, Loaders, Linkers, Macros, Compilers. software tools, Text editors, Interpreters and program generators, Debug Monitors, Programming environment.

Compiler: Brief overview of compilation process, Incremental compiler, Assembler: Problem statement, single phase and two phase assembler, symbol table; Loader schemes, compile and go Loader, general loader schemes, absolute loader, Subroutine linkage, Reallocating loader, Directlinkage Loader, Binders, Linking loader, overlays.

MODULE-II

Macro language and macro-processor, macro instructions, features of macro facility, macro instruction arguments, conditional macro expansion, macro calls with macro instruction defining macros.
Theoretical Concept of Unix Operating System: Basic features of operating system, File structure: CPU scheduling; Memory management: swapping, demand paging; file system: block and fragments, inodes, directory structure; User to user communication.

MODULE-III

Getting Started with Unix: User names and groups, logging in; Format of Unix commands; Changing your password; Characters with special meaning; Unix documentation; Files and directories; Current directory, looking at the directory contents, absolute and relative pathnames, some Unix directories and files; Looking at the file contents; File permissions; basic operation on files; changing permission modes; Standard files, standard output; Standard input, standard error; filters and pipelines; Processes; finding out about processes; Stopping background process; Unix editor vi. Test Manipulation: Inspecting files; File statistics; Searching for patterns; Comparing files; Operating on files; Printing files; Rearranging files; Sorting files; Splitting files; Translating characters; AWK utility. Shell Programming: Programming in the Borne and C-Shell; Wild cards; Simple shell programs; Shell variables; Shell programming constructs; interactive shell scripts; Advanced features.

Text Books:

1. Systems Programming by Donovan, TMH.
2. The unix programming environment by Brain Kernighen & Rob Pike, 1984, PHI & Rob Pike.
3. Design of the Unix operating system by Maurich Bach, 1986, PHI.
4. Introduction to UNIX and LINUX by John Muster, 2003, TMH.

Reference Books:

1. Advanced Unix programmer's Guide by Stephen Prato, BPB
2. Unix- Concept and applications by Sumitabha Das, 2002, TMH

Course Outcomes: After the completion of the course the student will be able to:

CO1	To understand the basics of system programs like editors, compiler, assembler, linker, loader, interpreter and debugger
CO2	Describe the various concepts of assemblers and microprocessors.
CO3	To understand the various phases of compiler and compare its working with assembler.
CO4	To understand how linker and loader create an executable program from an object module created by assembler and compiler
CO5	To know various editors and debugging techniques.